Proposed Solution Document

Date	18 October 2026
Team ID	Rahul-DS633
Project Name	CrimeVision AI - Intelligent Multimodal Crime Scene Reconstruction
Maximum Marks	5 Marks

1. Executive Solution Overview

CrimeVision AI is an intelligent multimodal forensic analysis and 3D crime scene reconstruction platform. It ingests visual evidence photos, acoustic 911/witness statements, and CCTV surveillance video, processing them through a unified pipeline of computer vision (RT-DETR/CLIP), speech recognition (Whisper), temporal event synchronization, and generative reasoning (Groq LLaMA-3.3).

2. Key Architectural Pillars

Pillar	Architecture & Model	Deliverable
1. Spatial 3D Recovery	RT-DETR detection + depth projection + Three.js canvas.	Interactive 3D room with classified evidence cubes.
2. Acoustic Intelligence	Whisper multilingual pipeline + Groq translation engine.	Live transcript & 4-language translation tabs.
3. CCTV Timeline	1.0 Hz video decomposition + BLIP captioning.	Second-by-second chronological incident log.
4. Multimodal Fusion	Cross-modal entity resolver & consistency checker.	Unified Case Dossier eliminating contradictions.
5. AI Chatbot	Groq LLaMA-3.3-70B grounded dialogue agent.	Natural language forensic Q&A; on evidence.
6. Court PDF Report	ReportLab Platypus + XML escape sanitation + SHA-256 seal.	Downloadable legal document ready for court.